

An Agent-Based LLM Framework for Inflation Forecasting: Evidence from Federal Reserve Communications

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Abstract

I focus on large language model (LLM)-based agents whether it can improve real-time inflation forecasting by incorporating both macroeconomic indicators and textual information contained in Federal Reserve communications. The study contributes new evidence using an agent-based expectation framework. I use monthly U.S. data from 2021-2025 and construct a real-time information set consisting of inflation, unemployment, industrial production, interest rates, and textual information extracted from Federal Open Market Committee (FOMC) statements. I develop a structural framework in which households, firms, and policymakers process both numerical and textual information when forming inflation expectations. Empirically, Federal Reserve communications are transformed into quantitative measures using FinBERT-based sentiment analysis and integrated into an inflation-forecasting model.¹ Forecast performance is evaluated against standard random-walk and autoregressive benchmarks. Preliminary results indicate that communication-based forecasts improve predictive accuracy. The FinBERT-based forecasting model achieves an RMSE of 2.61 and an MAE of 1.84, outperforming both random-walk forecasts (RMSE = 3.67, MAE = 2.41) and autoregressive benchmarks (RMSE = 3.66, MAE = 2.45). The results signal that expectation-relevant information embedded in policy communications can improve real-time inflation forecasting. More broadly, the findings provide preliminary support for modeling expectations as information-processing agents rather than relying exclusively on conventional time-series dynamics.

Keywords: Inflation Forecasting; Inflation Expectations; Large Language Models; Federal Reserve Communications; Text-as-Data; Central Bank Communication; Monetary Policy; Real-Time Data.

JEL Codes: C53; C55; D84; E31; E37; E52; E58.

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¹FinBERT is used as an empirical device for extracting information embedded in Federal Reserve communications. The objective is not to measure sentiment per se, but rather to recover signals that may reflect policymakers' assessments of future inflationary conditions and monetary-policy risks.

Introduction

Inflation expectations anchor modern macroeconomics and shape monetary policy. Expectations influence household consumption and saving decisions, firm pricing behavior, wage negotiations, investment planning, and ultimately the transmission of monetary policy throughout the economy (Lucas, 1972; Woodford, 2003; Gali, 2015). Despite decades of research, forecasting inflation remains difficult. Traditional forecasting approaches, including autoregressive models, Phillips-curve specifications, vector autoregressions, and dynamic stochastic general equilibrium models, often struggle during periods of rapid economic change and heightened uncertainty (Stock and Watson, 2007; Atkeson and Ohanian, 2001; Faust and Wright, 2013). A growing literature argues that economic agents do not possess perfect information and instead form expectations using limited, incomplete, and evolving information sets (Sims, 2003; Mankiw and Reis, 2002; Evans and Honkapohja, 2001).

One potentially valuable source of information is central-bank communication. Monetary policy operates not only through interest-rate adjustments but also through guidance, statements, speeches, and policy narratives that shape expectations regarding future inflation and economic activity (Bernanke, 2004; Blinder, 1998). In the United States, communications released by the Federal Open Market Committee (FOMC) provide real-time information regarding policymakers’ assessments of inflationary pressures, labor-market conditions, economic risks, and future policy intentions. Recent advances in text-as-data methods have enabled researchers to quantify the information content into forecasting models (Hansen et al., 2018; Shapiro et al., 2022). A key challenge in using policy communications for forecasting is determining whether textual information contains predictive content beyond publicly available macroeconomic indicators. Federal Reserve communications are not generated independently of economic data; rather, they represent a public signal of a substantially larger information set available to policymakers. Let

$$\Omega_t^{Fed} = \pi_t, PCE_t, u_t, FC_t, RR_t, SF_t, MI_t, BB_t, \dots, \quad (1)$$

where π_t denotes inflation, PCE_t personal consumption expenditures inflation, u_t labor-market conditions, FC_t financial conditions, RR_t regional economic reports, SF_t staff forecasts, MI_t market intelligence, and BB_t information contained in the Beige Book. FOMC statements can therefore be interpreted as compressed summaries of policymakers’ assessments of this broader information set. Consequently, textual communications may contain expectation-relevant information regarding future economic conditions and policy responses that is not fully captured by contemporaneous macroeconomic variables alone (Bernanke, 2004; Blinder, 1998; Hansen et al., 2018).

Another research suggests that language models are capable of generating economically meaningful forecasts and expectation updates using information similar to that observed by households, firms, and policymakers (Korinek, 2024; Li et al., 2024). This raises the possibility of modeling expectations as information-processing agents that learn from real-time macroeconomic conditions and policy communications.²

This paper asks a simple but important question: *Can an agent-based expectation framework that incorporates Federal Reserve communications improve real-time inflation forecasting?* To address this question, I develop a structural framework in which households, firms, and policymakers form expectations using real-time macroeconomic information and textual information extracted from Federal Reserve communications. The framework combines insights from expectation-formation theory, text-as-data methods, and recent developments in language-based artificial intelligence.

I construct a real-time information set, using monthly U.S. data from 2021-2025, consisting of inflation, unemployment, industrial production, the federal funds rate, and textual information extracted from FOMC statements. Preliminary forecasting suggest that Federal Reserve communications contain forward-looking information about inflation dynamics that is not fully captured by conventional macroeconomic indicators alone.

This paper contributes to the literature in three ways. First, it develops an agent-based expectation framework that integrates households, firms, and policymakers within a unified inflation-forecasting environment. Unlike conventional macroeconomic models that impose rational expectations, the framework allows agents to form expectations using real-time quantitative and textual information, drawing on insights from rational inattention, adaptive learning, and information-frictions literatures (Sims, 2003; Mankiw and Reis, 2002; Evans and Honkapohja, 2001). Second, the paper incorporates textual information from Federal Reserve communications into the expectation-formation process, extending the growing literature on central-bank communication and text-as-data methods (Hansen et al., 2018; Shapiro et al., 2022). Third, the analysis provides a framework through which advances in large language models can be incorporated into real-time macroeconomic forecasting and policy analysis, bridging recent developments in artificial intelligence with traditional expectation-formation research (Korinek, 2024; Li et al., 2024).

²The structural framework developed in this paper is motivated by recent advances in large language models and agent-based expectation formation. The empirical implementation uses FinBERT-based sentiment extraction from Federal Reserve communications as a tractable approximation to the broader LLM-agent framework.

Literature Review

This paper relates to four strands of literature: inflation forecasting, expectation formation, central-bank communication and text-as-data, and the emerging literature on large language models in economics.

i. Inflation Forecasting

A large literature examines the determinants and predictability of inflation. [Atkeson and Ohanian \(2001\)](#) argue that simple forecasting models often perform as well as more sophisticated alternatives, while [Stock and Watson \(2007\)](#) document the instability of inflation forecasting relationships over time. Recently, [Faust and Wright \(2013\)](#) emphasize the difficulty of producing accurate inflation forecasts even with large information sets.

ii. Expectation Formation

Modern macroeconomic theory places expectations at the center of economic decision making. Rational-expectations models assume that agents efficiently process all available information when forming forecasts ([Lucas, 1972](#); [Muth, 1961](#)). However, a growing body of research argues that information-processing frictions play an important role in shaping expectations. Rational inattention models emphasize the costs of acquiring and processing information ([Sims, 2003](#)), while sticky-information models suggest that agents update information infrequently ([Mankiw and Reis, 2002](#)). Adaptive-learning frameworks further allow expectations to evolve as agents learn about the economy over time ([Evans and Honkapohja, 2001](#)). These approaches imply that expectation formation depends critically on the information available to economic agents and the mechanisms through which information is processed.

iii. Central-Bank Communication and Text-as-Data

Monetary policy operates through both policy actions and communication. [Bernanke \(2004\)](#) and [Blinder \(1998\)](#) argue that communication has become an increasingly important policy instrument. Advances in text-as-data methods have enabled researchers to quantify the informational content of policy communications. For example, [Hansen et al. \(2018\)](#) demonstrate that textual analysis of Federal Reserve communications provides useful information about policy decisions and economic outlooks. Similarly, [Shapiro et al. \(2022\)](#) show that textual measures can capture economically meaningful signals related to inflation and macroeconomic conditions.

iv. Large Language Models and Economic Forecasting

On the other hand, recent advances in artificial intelligence have generated interest in the use of large language models for economic analysis. Unlike traditional forecasting methods that rely primarily on structured numerical data, large language models are capable of generating economically meaningful forecasts, summarizing policy information, and replicating aspects of human expectation formation (Korinek, 2024; Li et al., 2024). While this literature remains in its early stages, it raises important questions regarding how language models can be integrated into economic forecasting and policy analysis. This paper contributes to this emerging area by developing an agent-based framework in which households, firms, and policymakers form expectations using real-time macroeconomic information and Federal Reserve communications.

Overall, the existing literature highlights the importance of expectations, information processing, and policy communication for macroeconomic outcomes. However, relatively little research has examined how language-model-based agents can integrate real-time quantitative and textual information within a unified forecasting framework. This paper seeks to address that gap.

Road Map. The remainder of the paper proceeds as follows. Section I describes the data sources and construction of the real-time information set. Section II develops the agent-based theoretical framework and introduces the LLM expectation operators for households, firms, and policymakers. Section III presents the empirical forecasting methodology and benchmark models. Section IV reports the forecasting results and evaluates model performance. Section V discusses the findings and their implications for expectation formation and monetary-policy communication. Section VI concludes and outlines directions for future research.

I. Data

The sample consists of monthly observations from January 2021 through December 2025. The period covers the post-pandemic inflation surge, the subsequent monetary tightening cycle, and the transition toward lower inflation, providing substantial variation in both macroeconomic conditions and Federal Reserve communications.

i. Macroeconomic Data

Macroeconomic variables are obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. FRED provides publicly available and widely used macroeconomic time-series data for economic research and

policy analysis ([Federal Reserve Bank of St. Louis, 2024](#)). The baseline information set includes four macroeconomic indicators:

1. Consumer Price Index (CPI)
2. Unemployment Rate
3. Federal Funds Rate
4. Industrial Production Index

Consumer prices are measured using the Consumer Price Index for All Urban Consumers (CPIAUCSL), published by the U.S. Bureau of Labor Statistics. CPI inflation serves as the primary outcome variable because it is one of the most widely monitored indicators of aggregate price dynamics and plays a central role in monetary policy deliberations ([Stock and Watson, 2007](#)). Labor-market conditions are measured using the civilian unemployment rate (UNRATE). Unemployment provides information regarding labor-market slack and enters many traditional inflation forecasting frameworks through Phillips-curve relationships ([Phillips, 1958](#); [Gali, 2015](#)). Monetary policy conditions are measured using the effective federal funds rate (FEDFUNDS), which represents the Federal Reserve’s primary policy instrument ([Taylor, 1993](#)). Changes in the federal funds rate affect aggregate demand, financial conditions, and inflation expectations. Real economic activity is measured using the Industrial Production Index (INDPRO), published by the Board of Governors of the Federal Reserve System. Industrial production is frequently used as a high-frequency indicator of aggregate economic activity and business-cycle conditions ([Bernanke et al., 1997](#)).

Monthly industrial production growth is constructed as

$$g_t^{IP} = 100 [\log(INDPRO_t) - \log(INDPRO_{t-1})]. \quad (2)$$

Monthly inflation is calculated using the logarithmic growth rate of the CPI ³:

$$\pi_t = 1200 [\log(CPI_t) - \log(CPI_{t-1})], \quad (3)$$

ii. Federal Reserve Communication Data

Textual information is obtained from official Federal Open Market Committee (FOMC) policy statements published on the Federal Reserve’s website. FOMC statements are released immediately following scheduled policy meetings and summarize policymakers’

³where the factor 1200 converts monthly inflation into annualized percentage terms.

assessments of inflation, labor-market conditions, economic growth, financial conditions, and future policy risks. The communication literature argues that central-bank statements contain economically meaningful information beyond policy-rate decisions themselves (Bernanke, 2004; Blinder, 1998; Hansen et al., 2018). Text preprocessing consists of removing HTML markup, standardizing whitespace, and converting documents into machine-readable text. The resulting corpus forms the textual component of the real-time information set.

iii. Text-Based Information Extraction

To quantify the information contained in Federal Reserve communications, the analysis employs Financial Bidirectional Encoder Representations from Transformers (FinBERT), a transformer-based language model trained on financial text (Araci, 2019).

For each FOMC statement, FinBERT generates probabilities associated with positive, negative, and neutral sentiment:

$$S_t = (s_t^{pos}, s_t^{neg}, s_t^{neu}), \quad (4)$$

where

$$s_t^{pos} + s_t^{neg} + s_t^{neu} = 1. \quad (5)$$

A sentiment-balance measure is constructed as

$$SB_t = s_t^{pos} - s_t^{neg}. \quad (6)$$

Positive values indicate relatively optimistic communication, whereas negative values indicate relatively inflation-concerned communication. These text-derived measures provide quantitative representations of policy communication and serve as inputs into the expectation-formation process.

iv. Real-Time Information Set

The agent-based framework assumes that economic agents observe a common information set consisting of both numerical and textual information:

$$\mathcal{I}_t = \{\pi_t, u_t, i_t, g_t^{IP}, SB_t, z_t\}, \quad (7)$$

where π_t denotes inflation, u_t the unemployment rate, i_t the federal funds rate, g_t^{IP} industrial-production growth, SB_t communication-based expectation signal extracted from

Federal Reserve communications, and z_t the underlying FOMC statement text.

This information set approximates the real-time environment in which households, firms, and policymakers form expectations regarding future inflation.

v. Summary Statistics

Table 1: Summary Statistics

	Obs	Mean	Std. Dev.	Min	P25	Median	P75	Max
inflation	57	4.484	3.364	-0.502	2.307	3.694	5.789	14.979
inflation_next	57	4.485	3.364	-0.502	2.307	3.694	5.789	14.979
UNRATE	57	4.172	0.768	3.400	3.600	3.900	4.200	6.400
FEDFUNDS	57	3.194	2.183	0.060	0.200	4.330	5.120	5.330
output_growth	57	0.696	9.823	-40.321	-3.176	0.583	5.451	34.279
finbert_positive	57	0.185	0.151	0.059	0.101	0.129	0.161	0.622
finbert_negative	57	0.129	0.091	0.013	0.062	0.104	0.163	0.421
finbert_neutral	57	0.686	0.148	0.346	0.583	0.731	0.782	0.876
sentiment_balance	57	0.055	0.201	-0.362	-0.031	0.005	0.057	0.589

Table 1 reports summary statistics for all macroeconomic and text-derived variables used in the analysis. The sample exhibits substantial variation in inflation, policy rates, and communication sentiment, reflecting the evolving macroeconomic environment during the post-pandemic inflation episode and subsequent monetary-policy tightening cycle.

vi. Descriptive Evidence and Visualization

The descriptive evidence provides preliminary insights into the macroeconomic environment faced by households, firms, and policymakers and motivates the construction of the agent-based forecasting framework.

a. Inflation and Monetary Policy Conditions

Figure 1 plots annualized monthly inflation and the effective federal funds rate. The figure illustrates the post-pandemic inflation surge that began in 2021 and intensified throughout 2022. In response, the Federal Reserve implemented one of the most rapid monetary tightening cycles in recent decades, increasing the federal funds rate from near-zero levels to above five percent.

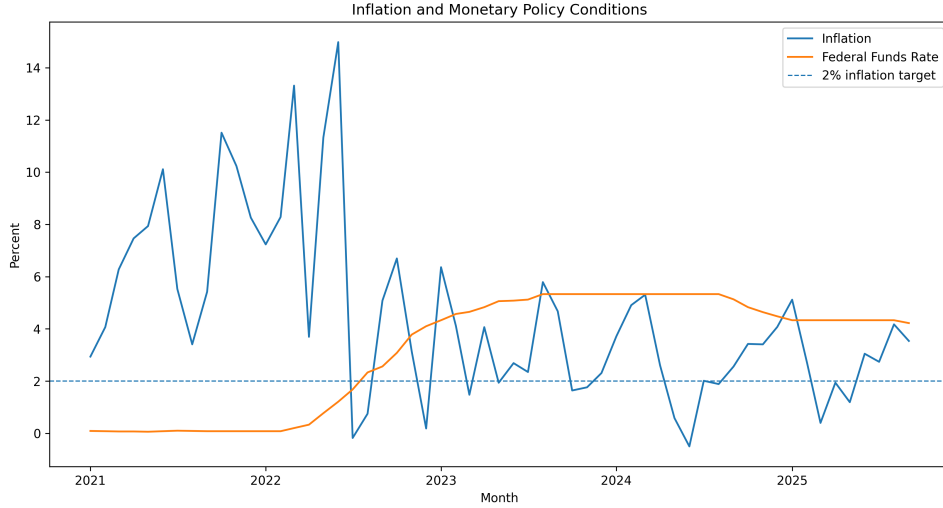


Figure 1: Inflation increased substantially during 2021–2022 and was followed by a rapid monetary-policy tightening cycle.

The figure highlights the close interaction between inflation and monetary policy, which is consistent with modern monetary theory, and emphasizes the role of interest-rate policy in stabilizing inflation and economic activity (Taylor, 1993; Woodford, 2003; Gali, 2015).

b. Sentiment in Federal Reserve Communications

Figure 2 reports the sentiment-balance measure constructed from FOMC statements using FinBERT. The ability of central-bank communications to influence expectations is widely documented in the monetary-policy literature (Bernanke, 2004; Blinder, 1998; Hansen et al., 2018). Consequently, textual measures may contain information relevant for forecasting future inflation.

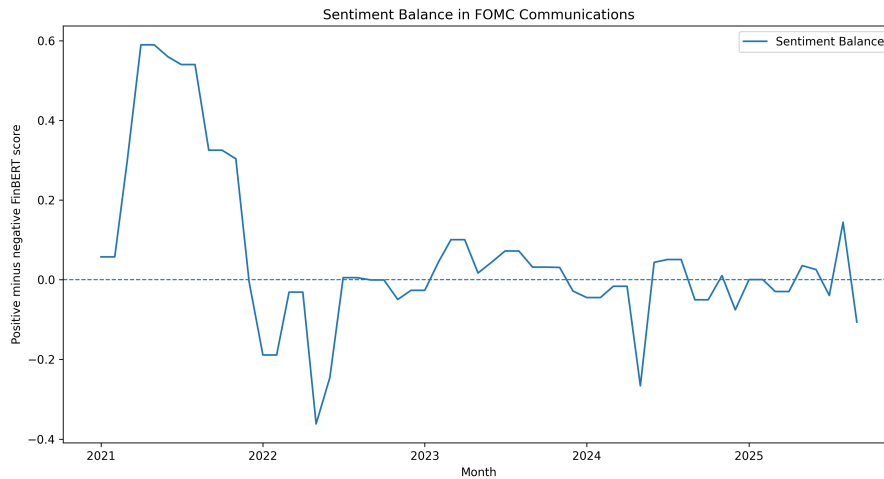


Figure 2: Sentiment balance extracted from FOMC statements using FinBERT.

c. Hawkish and Dovish Communication

Figure 3 decomposes Federal Reserve communications into hawkish and dovish language components. Hawkish language is generally associated with concerns regarding inflationary pressures and tighter monetary policy, whereas dovish language emphasizes economic support and accommodative policy conditions. The patterns are consistent with evidence that central-bank communications evolve systematically in response to macroeconomic conditions and policy objectives (Blinder, 1998; Hansen et al., 2018).⁴

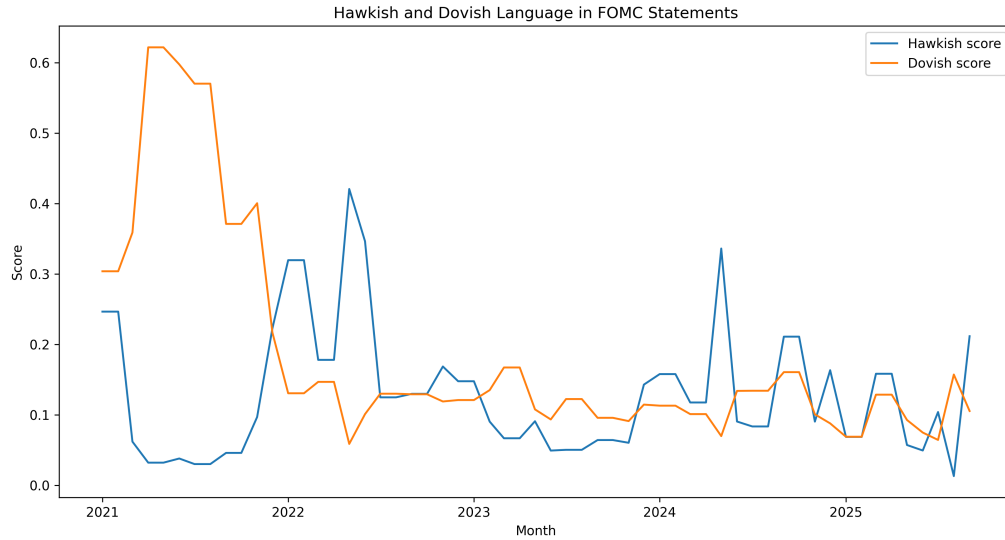


Figure 3: Hawkish and dovish language extracted from FOMC statements. Dovish language dominates during the early post-pandemic period, while hawkish language rises during the 2022 monetary-tightening cycle. The convergence of the two measures after 2023 suggests a gradual normalization of policy communication.

Taken together, Figures 1- 3 reveal three important features of the sample period. First, inflation and monetary policy experienced substantial variation, providing a challenging environment for forecasting models. Second, Federal Reserve communications evolved significantly as policymakers responded to changing economic conditions. Third, textual information extracted from FOMC statements exhibits meaningful time variation that may contain information not fully captured by traditional macroeconomic indicators.

These observations motivate the structural framework developed in the next section, where households, firms, and policymakers form expectations using both quantitative macroeconomic information and qualitative information contained in Federal Reserve communications.

II. Structural Framework

i) The LLM Economy

The economy consists of households, firms, a monetary authority, and an LLM-based expectation-formation technology. The model follows a standard New Keynesian structure with intertemporal household optimization, monopolistically competitive firms, and an interest-rate rule (Woodford, 2003; Gali, 2015). The innovation is that agents form expectations using real-time macroeconomic and textual information processed through large language model agents, rather than imposing rational expectations.

At time t , recall, agents observe the real-time information set

$$\mathcal{I}_t = \{\pi_t, u_t, i_t, g_t^{IP}, SB_t, z_t\}, \quad (8)$$

The information set $\mathcal{I}_t$ represents the publicly observable information available to economic agents when forming expectations regarding future inflation and economic activity. In the empirical implementation, aggregate output is proxied by industrial production growth because monthly GDP data are unavailable. The communication variables SB_t and z_t capture information contained in Federal Reserve statements and are interpreted as observable signals of policymakers' assessments regarding future economic conditions rather than simple restatements of contemporaneous macroeconomic data. The LLM expectation operators map the economy-wide information set into agent-specific forecasts. Household expectations are generated according to

$$E_{i,t}^{LLM}(x_{t+1}) = f_H(\mathcal{I}_t, \chi_i; \theta_H), \quad (9)$$

Firm expectations are

$$E_{j,t}^{LLM}(x_{t+1}) = f_F(\mathcal{I}_t, \psi_j; \theta_F), \quad (10)$$

and Federal Reserve expectations are

$$E_{Fed,t}^{LLM}(x_{t+1}) = f_{Fed}(\mathcal{I}_t, z_t^{Fed}; \theta_{Fed}). \quad (11)$$

Household expectations are generated by $f_H(\cdot)$, firm expectations by $f_F(\cdot)$, and Federal Reserve expectations by $f_{Fed}(\cdot)$. The vectors χ_i and ψ_j capture household-specific and firm-specific characteristics, respectively. The parameters θ_H , θ_F , and θ_{Fed} govern the behavior of the corresponding LLM expectation operators. Consequently, households, firms, and the Federal Reserve may process the same aggregate information set differently, generating heterogeneous expectations regarding future inflation, output, and other

macroeconomic variables.

The key departure from rational expectations is

$$E_t^{LLM}(\cdot) \neq E_t^{RE}(\cdot). \quad (12)$$

The framework does not impose rational expectations. Instead, agents form expectations using observable macroeconomic information, textual signals, and agent-specific information-processing rules. Consequently, expectation formation may deviate from the fully informed rational-expectations benchmark. This interpretation is consistent with the literature on central-bank communication, forward guidance, and expectation management (Bernanke, 2004; Blinder, 1998; Hansen et al., 2018).

This assumption is motivated by rational inattention, sticky information, adaptive learning, and survey-expectations literatures (Sims, 2003; Mankiw and Reis, 2002; Evans and Honkapohja, 2001; Coibon and Gorodnichenko, 2015). The LLM does not replace economic optimization; it replaces the mechanism through which agents transform real-time information into expectations. Recent research suggests that large language models can process economic information, generate forecasts, and produce expectation updates that are informative for economic decision-making (Korinek, 2023, 2024; Horton, 2023).

ii. Federal Reserve LLM Agent

The Federal Reserve is modeled as a forward-looking policy agent that processes real-time macroeconomic and textual information, emphasizing that monetary policy operates not only through current policy actions but also expectations regarding future policy and economic conditions (Bernanke, 2004; Blinder, 1998; Woodford, 2005; Campbell et al., 2012).

The Federal Reserve is assumed to observe a broader information set Ω_t^{Fed} that includes the publicly observable macroeconomic information set $\mathcal{I}_t$ together with staff forecasts, regional economic reports, market intelligence, and other policy-relevant information:

$$\Omega_t^{Fed} = \{\mathcal{I}_t, \text{staff forecasts, regional reports, market intelligence, Beige Book information}\}. \quad (13)$$

Consequently,

$$\Omega_t^{Fed} \supset \mathcal{I}_t. \quad (14)$$

The communication component z_t^{Fed} denotes publicly released Federal Reserve commu-

communications, including FOMC statements used in the empirical analysis. These communications are interpreted as observable signals of policymakers' beliefs regarding future inflation, economic activity, and prospective policy actions.

The Federal Reserve forms expectations according to

$$E_{Fed,t}^{LLM}(x_{t+1}) = f_{Fed}(\Omega_t^{Fed}, z_t^{Fed}; \theta_{Fed}). \quad (15)$$

where θ_{Fed} denotes the parameterization of the Federal Reserve LLM agent.

Expected inflation is

$$E_{Fed,t}^{LLM}(\pi_{t+1}) = f_{Fed}^{\pi}(\Omega_t^{Fed}, z_t^{Fed}). \quad (16)$$

and expected output is

$$E_{Fed,t}^{LLM}(y_{t+1}) = f_{Fed}^y(\Omega_t^{Fed}, z_t^{Fed}). \quad (17)$$

The Federal Reserve expectation operator is interpreted as a computational representation of policy-makers processing large volumes of quantitative and qualitative information when assessing future inflation and economic activity. Rather than imposing model-consistent rational expectations, the framework allows policy expectations to be formed from real-time macroeconomic indicators and policy communications, consistent with modern views of information frictions and expectation management (Woodford, 2005; Campbell et al., 2012).

iii. Households

There is a continuum of households indexed by $i \in [0, 1]$. Household i chooses consumption, labor supply, and nominal bond holdings:

$$\{C_{i,t}, N_{i,t}, B_{i,t+1}\}. \quad (18)$$

Households are modeled as forward-looking agents that smooth consumption over time while supplying labor and allocating savings through one-period nominal bonds. This structure follows the standard intertemporal consumption and labor supply framework of Hall (1978), Woodford (2003), and Gali (2015).

Household i maximizes expected lifetime utility:

$$\max_{\{C_{i,t}, N_{i,t}, B_{i,t+1}\}} E_{i,t}^{LLM} \sum_{s=0}^{\infty} \beta^s \left[\frac{C_{i,t+s}^{1-\sigma}}{1-\sigma} - \frac{\varphi N_{i,t+s}^{1+\eta}}{1+\eta} \right], \quad (19)$$

where $\beta \in (0, 1)$ is the discount factor, $\sigma > 0$ is the coefficient of relative risk aversion, and $\eta > 0$ governs the Frisch elasticity of labor supply.

The constant-relative-risk-aversion (CRRA) specification is standard in modern macroeconomics because it yields tractable intertemporal consumption decisions while allowing risk aversion and intertemporal substitution to be governed by a single parameter (Hall, 1978; Sargent and Ljungqvist, 2018).

The household budget constraint is,

$$P_t C_{i,t} + B_{i,t+1} = W_t N_{i,t} + (1 + i_{t-1}) B_{i,t}, \quad (20)$$

where P_t is the aggregate price level, W_t is the nominal wage, and $B_{i,t+1}$ denotes one-period nominal bond holdings. Bonds purchased in period t mature in period $t + 1$ and pay the gross nominal return $(1 + i_t)$.

Households receive income from labor and bond returns. Capital ownership and investment decisions are not modeled at the household level because the focus of the paper is inflation expectations, monetary-policy transmission, and real-time forecasting rather than wealth accumulation and capital ownership.

The first-order condition for bond holdings yields the consumption Euler equation

$$C_{i,t}^{-\sigma} = \beta E_{H,i,t}^{LLM} \left[C_{i,t+1}^{-\sigma} \frac{1 + i_t}{1 + \pi_{t+1}} \right]. \quad (21)$$

This equation governs optimal intertemporal consumption smoothing and links household decisions to monetary policy through the expected real return on savings (Hall, 1978; Woodford, 2003; Gali, 2015).

The household optimization problem is completed by the transversality condition:

$$\lim_{T \rightarrow \infty} E_{i,t}^{LLM} \left[\beta^T \frac{B_{i,t+T}}{\prod_{s=0}^{T-1} (1 + i_{t+s})} \right] = 0. \quad (22)$$

This condition rules out Ponzi schemes and ensures that households cannot finance consumption indefinitely through perpetual debt accumulation (Sargent and Ljungqvist, 2018; Woodford, 2003).

The expected gross real return is,

$$R_{i,t}^e = \frac{1 + i_t}{1 + E_{H,i,t}^{LLM}(\pi_{t+1})}. \quad (23)$$

Using a log-linear approximation, the expected real interest rate is

$$r_{i,t}^e \approx i_t - E_{H,i,t}^{LLM}(\pi_{t+1}). \quad (24)$$

Consequently, expected inflation affects household consumption decisions through its effect on the expected real return to saving.

The household labor-supply condition is

$$\varphi N_{i,t}^\eta = C_{i,t}^{-\sigma} \frac{W_t}{P_t}. \quad (25)$$

This condition equates the marginal disutility of supplying labor with the marginal utility value of labor income. Therefore, labor supply depends on the real wage and the marginal utility of consumption (Woodford, 2003; Gali, 2015).

Together, the budget constraint, Euler equation, and labor-supply condition imply that household behavior depends on expected inflation, expected real interest rates, labor income, and real wages. Household expectations are heterogeneous because agents may process information differently. Aggregate expectations are obtained by integrating over the continuum of households.

a. Aggregate Household Inflation Expectations

Households form inflation expectations using an LLM-based expectation operator:

$$E_{H,i,t}^{LLM}(\pi_{t+1}) = f_H^\pi(\mathcal{I}_t, \chi_i; \theta_H). \quad (26)$$

Because households are modeled as a continuum, aggregate household inflation expectations are

$$\bar{\pi}_{H,t+1}^{LLM} = \int_0^1 E_{H,i,t}^{LLM}(\pi_{t+1}) di. \quad (27)$$

Heterogeneity arises because households may differ in their information-processing characteristics χ_i , causing agents exposed to the same aggregate information set to form different inflation expectations. In the empirical implementation, household, firm, and Federal Reserve expectation operators are approximated using prompt-based language agents that receive identical macroeconomic information but differ in their roles, objectives, and information-processing instructions. The resulting agent-specific forecasts provide empirical counterparts to the theoretical expectation operators.

iv. Firms

There is a continuum of monopolistically competitive firms indexed by $j \in [0, 1]$. Firm j produces differentiated output using capital and labor:

$$Y_{j,t} = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}, \quad 0 < \alpha < 1. \quad (28)$$

This Cobb-Douglas production function follows the standard macroeconomic production framework of [Cobb and Douglas \(1928\)](#), [King et al. \(1988\)](#), and [Gali \(2015\)](#). The production environment is dynamic because aggregate productivity evolves over time:

$$\log A_t = \rho_A \log A_{t-1} + \varepsilon_{A,t}, \quad |\rho_A| < 1. \quad (29)$$

Capital accumulation is defined at the aggregate level as

$$K_{t+1} = (1 - \delta)K_t + I_t, \quad (30)$$

where I_t denotes aggregate investment and δ is the depreciation rate. This structure follows standard dynamic macroeconomic models with time-varying productivity and capital accumulation ([Woodford, 2003](#); [Gali, 2015](#); [Sargent and Ljungqvist, 2018](#)).

Demand for differentiated goods follows the Dixit–Stiglitz monopolistic-competition framework ([Dixit and Stiglitz, 1977](#)). Demand for firm j is

$$Y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t, \quad \varepsilon > 1. \quad (31)$$

Firm marginal cost is

$$MC_{j,t} = \frac{1}{A_t} \left(\frac{W_t}{1 - \alpha} \right)^{1-\alpha} \left(\frac{R_t^K}{\alpha} \right)^\alpha. \quad (32)$$

Here R_t^K denotes the rental price of capital and W_t denotes the nominal wage. Marginal cost is obtained from the firm's cost-minimization problem subject to the Cobb–Douglas production technology ([Gali, 2015](#)).

Firm expectations are generated by

$$E_{j,t}^{LLM}(x_{t+1}) = f_F(\mathcal{I}_t, \bar{\pi}_{H,t+1}^{LLM}, \psi_j; \theta_F), \quad (33)$$

where ψ_j captures firm-specific characteristics or prompt profiles and θ_F denotes firm-side LLM parameters. Firms are assumed to incorporate aggregate household inflation expectations when forming pricing decisions because expected consumer demand and inflation perceptions influence future pricing behavior.

Following the staggered-price framework of [Calvo \(1983\)](#), firms adjust desired prices in response to expected inflation, expected aggregate demand, and expected marginal costs. The desired price adjustment is

$$\Delta p_{j,t+1}^* = \phi_1 E_{j,t}^{LLM}(\pi_{t+1}) + \phi_2 E_{j,t}^{LLM}(Y_{t+1}) + \phi_3 E_{j,t}^{LLM}(MC_{j,t+1}) + \nu_{j,t+1}. \quad (34)$$

Aggregate firm-implied inflation is

$$\bar{\pi}_{F,t+1}^{LLM} = \int_0^1 \Delta p_{j,t+1}^* dj. \quad (35)$$

a. Firm Transversality Condition

$$\lim_{T \rightarrow \infty} E_{j,t}^{LLM} [\beta^T K_{j,t+T}] = 0. \quad (36)$$

v. Market Clearing Conditions

a. Aggregate Household Quantities

$$C_t = \int_0^1 C_{i,t} di, \quad (37)$$

b. Goods Market

The goods-market condition

$$Y_t = C_t + I_t + G_t. \quad (38)$$

Aggregate capital evolves according to

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (39)$$

c. Labor Market

Aggregate firm labor demand is

$$N_t^d = \int_0^1 N_{j,t}^d dj. \quad (40)$$

Labor-market clearing requires

$$N_t^s = N_t^d. \quad (41)$$

d. Bond Market

In the baseline specification, bonds are assumed to be in zero net supply. Therefore bond-market clearing requires

$$\int_0^1 B_{i,t+1} di = 0. \quad (42)$$

e. Capital Market

Aggregate capital demand is

$$K_t = \int_0^1 K_{j,t} dj. \quad (43)$$

Aggregate capital evolves according to

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (44)$$

f. Price Index

The firm's pricing problem follows the staggered-price framework of [Calvo \(1983\)](#):

$$P_t = \left[\int_0^1 P_{j,t}^{1-\varepsilon} dj \right]^{\frac{1}{1-\varepsilon}}. \quad (45)$$

This CES price index follows the standard monopolistic-competition framework ([Dixit and Stiglitz, 1977](#); [Calvo, 1983](#); [Woodford, 2003](#); [Gali, 2015](#)).

Final output is a CES aggregate of differentiated goods following [Dixit and Stiglitz \(1977\)](#):

$$Y_t = \left[\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (46)$$

Gross inflation is

$$1 + \pi_t = \frac{P_t}{P_{t-1}}. \quad (47)$$

Log inflation is

$$\pi_t = \log P_t - \log P_{t-1}. \quad (48)$$

vi. LLM-Based Inflation Forecast

The model-implied inflation forecast combines household, firm, and Federal Reserve inflation expectations:

$$\hat{\pi}_{t+1}^{LLM} = \omega_H \bar{\pi}_{H,t+1}^{LLM} + \omega_F \bar{\pi}_{F,t+1}^{LLM} + \omega_{Fed} E_{Fed,t}^{LLM}(\pi_{t+1}), \quad (49)$$

subject to

$$\omega_H + \omega_F + \omega_{Fed} = 1, \quad \omega_H, \omega_F, \omega_{Fed} \geq 0. \quad (50)$$

The weights determine the relative contribution of household expectations, firm pricing expectations, and Federal Reserve expectations to the aggregate inflation forecast.

vii. Structural Equilibrium

An LLM-agent structural equilibrium consists of household allocations $\{C_{i,t}, N_{i,t}, B_{i,t+1}\}_{i \in [0,1]}$, firm allocations $\{Y_{j,t}, N_{j,t}, K_{j,t}, P_{j,t}\}_{j \in [0,1]}$, aggregate variables $\{C_t, I_t, K_t, N_t, Y_t, P_t, \pi_t, i_t\}$, and LLM-generated expectations such that:

1. households satisfy the budget constraint, bond Euler equation, and labor-supply condition;
2. firms satisfy production, demand, marginal-cost, and pricing conditions;
3. goods-market clearing holds:

$$Y_t = C_t + I_t + G_t;$$

4. capital accumulation holds:

$$K_{t+1} = (1 - \delta)K_t + I_t;$$

5. labor-market clearing holds:

$$N_t^s = N_t^d;$$

6. bond-market clearing holds:

$$\int_0^1 B_{i,t+1} di = 0;$$

7. aggregate output satisfies:

$$Y_t = \int_0^1 Y_{j,t} dj;$$

8. the aggregate price index satisfies:

$$P_t = \left[\int_0^1 P_{j,t}^{1-\varepsilon} dj \right]^{\frac{1}{1-\varepsilon}};$$

9. The monetary authority sets the nominal policy rate according to a Taylor-type rule (Taylor, 1993);
10. expectations are generated by LLM agents using $\mathcal{I}_t$;
11. the model-implied inflation forecast satisfies:

$$\hat{\pi}_{t+1}^{LLM} = \omega_H \bar{\pi}_{H,t+1}^{LLM} + \omega_F \bar{\pi}_{F,t+1}^{LLM} + \omega_{Fed} E_{Fed,t}^{LLM}(\pi_{t+1}).$$

III. Empirical Implementation

This section describes the empirical construction of the LLM-agent forecasting framework. The objective is to evaluate whether information extracted from Federal Reserve communications improves inflation forecasting relative to traditional benchmark models. The empirical implementation follows the structural framework developed above, where households, firms, and the Federal Reserve form expectations using both quantitative macroeconomic information and qualitative textual information.

i. Data Construction

The empirical analysis combines monthly macroeconomic indicators with textual information extracted from Federal Open Market Committee (FOMC) statements.

Macroeconomic variables are obtained from the Federal Reserve Economic Data (FRED) database. The baseline information set is

The empirical analysis operationalizes the theoretical information set $\mathcal{I}_t$ introduced in Section III using monthly U.S. macroeconomic indicators and textual information extracted from FOMC statements. where

- π_t denotes annualized monthly CPI inflation;
- u_t denotes the unemployment rate;
- i_t denotes the effective federal funds rate;
- g_t^{IP} denotes industrial-production growth;
- SB_t denotes the sentiment-balance measure extracted from FOMC statements;
- z_t denotes the underlying textual information.

Consumer price inflation is measured using the Consumer Price Index (CPI), which is widely used in inflation forecasting studies ([Atkeson and Ohanian, 2001](#)). Labor-market conditions are captured using the civilian unemployment rate, while monetary-policy conditions are measured using the effective federal funds rate. Industrial production growth serves as a proxy for real economic activity and aggregate demand conditions.

ii. Federal Reserve Text Corpus

The monetary-policy communication literature argues that central-bank statements influence expectations of households, firms, and financial markets through a forward-guidance channel ([Bernanke, 2004](#); [Blinder, 1998](#); [Hansen et al., 2018](#)). Consequently, textual information may contain predictive information about future inflation beyond conventional macroeconomic indicators. For each month, the most recent FOMC statement is matched to the macroeconomic data. This produces a real-time information set that mimics the information available to economic agents at the time expectations are formed.

iii. Text Processing and Sentiment Extraction

Textual information is transformed into quantitative indicators using FinBERT, a transformer-based language model specifically designed for financial and economic text analysis ([Araci, 2019](#)). For each FOMC statement, FinBERT generates probabilities for three sentiment classes:

$$P_t^+, \quad P_t^-, \quad P_t^0, \quad (51)$$

representing positive, negative, and neutral sentiment respectively.

Following the text-analysis literature, a sentiment-balance measure is constructed as

$$SB_t = P_t^+ - P_t^-. \quad (52)$$

In addition, hawkish and dovish policy terms are used to construct supplementary communication indicators:

$$H_t = \frac{\text{Number of Hawkish Terms}}{\text{Total Words}}, \quad (53)$$

$$D_t = \frac{\text{Number of Dovish Terms}}{\text{Total Words}}. \quad (54)$$

These measures provide a direct representation of the policy stance embedded within Federal Reserve communications.

iv. Empirical Mapping of Structural Agents

The empirical implementation should therefore be viewed as a reduced-form approximation to the broader agent-based framework. The communication measures extracted from FOMC statements are interpreted as observable signals of policymakers’ assessments regarding future inflation and economic conditions. These signals provide empirical proxies for the expectation-formation process described in the structural model.

v. Inflation Forecasting Model

The baseline forecasting specification is

$$\pi_{t+1} = \alpha + \beta_1\pi_t + \beta_2u_t + \beta_3i_t + \beta_4g_t^{IP} + \beta_5SB_t + \varepsilon_t. \quad (55)$$

The coefficient β_5 captures the incremental predictive content of Federal Reserve communications after controlling for conventional macroeconomic indicators. The baseline specification should be interpreted as a communication-augmented inflation forecasting model rather than a direct estimation of the structural LLM-agent economy.

vi. Benchmark Forecasting Models

Forecast performance is evaluated relative to two standard benchmarks widely used in the inflation-forecasting literature.

The first benchmark is a random-walk forecast:

$$\hat{\pi}_{t+1}^{RW} = \pi_t. \quad (56)$$

The second benchmark is an autoregressive model:

$$\pi_{t+1} = \alpha + \rho\pi_t + \varepsilon_t. \quad (57)$$

Random-walk and autoregressive models have historically proven difficult to outperform in inflation forecasting exercises ([Atkeson and Ohanian, 2001](#)).

vii. Forecast Evaluation

Forecast accuracy is evaluated using root mean squared error (RMSE) and mean absolute error (MAE):

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (\pi_t - \hat{\pi}_t)^2}, \quad (58)$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |\pi_t - \hat{\pi}_t|. \quad (59)$$

Lower values indicate superior forecasting performance. The central empirical question is whether the incorporation of Federal Reserve communication and LLM-based expectation proxies improves forecasting accuracy relative to traditional benchmark models.

IV. Results

The regression analysis is estimated using the full sample ($N = 57$). Forecast evaluation is conducted on the common sample available for all competing forecasting models ($N = 44$), ensuring that forecast accuracy comparisons are based on identical observations.

i. Inflation Forecasting Regression

Table 2 reports estimates from the baseline forecasting specification:

$$\pi_{t+1} = \alpha + \beta_1 \pi_t + \beta_2 u_t + \beta_3 i_t + \beta_4 g_t^{IP} + \beta_5 SB_t + \varepsilon_t. \quad (60)$$

The model explains approximately 44 percent of the variation in one-month-ahead inflation over the sample period.

Among the explanatory variables, the federal funds rate exhibits a statistically significant negative relationship with future inflation. A one-percentage-point increase in the federal funds rate is associated with approximately a 1.06 percentage-point decline in subsequent inflation, consistent with standard monetary-policy transmission mechanisms. By contrast, the sentiment-balance measure is not statistically significant in the linear forecasting specification. This result suggests that the information contained in Federal Reserve communications may overlap substantially with conventional macroeconomic indicators. Nevertheless, communication-based measures may still contribute to forecast accuracy when combined with broader information sets. Relative to the random-walk benchmark, the communication-based model reduces RMSE by approximately 28.9 percent and MAE by approximately 23.9 percent. Relative to the autoregressive benchmark, RMSE declines by approximately 28.7 percent and MAE declines by approximately 25.1 percent.

Table 2: Baseline Inflation Forecasting Regression

Variable	Inflation _{t+1}
Inflation	0.059 (0.143)
Unemployment Rate	-0.621 (0.680)
Federal Funds Rate	-1.059*** (0.256)
Output Growth	0.0002 (0.037)
Sentiment Balance	-0.058 (2.372)
Constant	10.190*** (3.526)
Observations	57
R^2	0.436
Adjusted R^2	0.381
Standard errors in parentheses	
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$	

ii. Forecast Performance

Table 3: Forecast Performance Comparison

Model	RMSE	MAE
FinBERT Sentiment	2.611	1.835
Random Walk	3.671	2.413
ARIMA	3.663	2.449

Table 3 compares the forecasting accuracy of the communication-based model with two benchmark forecasting models. The FinBERT-based communication model achieves an RMSE of 2.61 and an MAE of 1.84. By comparison, the random-walk benchmark produces an RMSE of 3.67 and an MAE of 2.41, while the ARIMA benchmark yields an RMSE of 3.66 and an MAE of 2.45. Relative to the random-walk benchmark, the communication-based model reduces RMSE by approximately 29 percent. Similar improvements are observed relative to the ARIMA benchmark.

The communication-based model produces lower forecast errors than both benchmark models. These results are consistent with the view that information extracted from Fed-

eral Reserve communications may contain useful signals for future inflation beyond those captured by simple time-series benchmarks.

iii. Forecast Improvement

Relative to the random-walk benchmark, the communication-based model reduces RMSE by approximately 28.9 percent and MAE by approximately 23.9 percent. Relative to the autoregressive benchmark, RMSE declines by approximately 28.7 percent and MAE declines by approximately 25.1 percent. These improvements indicate that information extracted from Federal Reserve communications contributes to forecasting accuracy beyond traditional benchmark models.

iv. Forecast Dynamics

Figure 4 compares actual inflation with forecasts generated by the communication-based model, the random-walk benchmark, and the autoregressive benchmark.

The communication-based forecast tracks medium-run inflation dynamics more closely than the benchmark models. Although no forecasting model perfectly captures large inflation spikes, the communication-based specification produces smoother and more accurate predictions during periods of elevated inflation volatility. These findings suggest that textual information extracted from Federal Reserve communications contains economically meaningful information regarding future inflation developments.

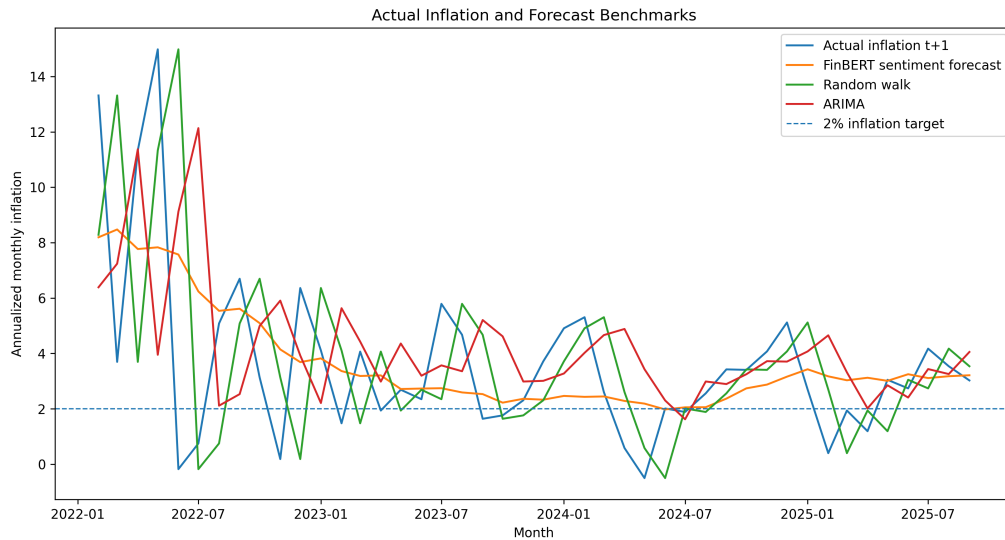


Figure 4: Actual inflation and forecast benchmarks. The communication-based forecast tracks medium-run inflation dynamics more closely than traditional benchmark models.

V. Discussion

This paper investigates whether information extracted from Federal Reserve communications can improve real-time inflation forecasting. The preliminary empirical results indicate that communication-based forecasting models outperform conventional benchmark specifications, including random-walk and autoregressive forecasts. In particular, the FinBERT-based communication measure achieves substantially lower RMSE and MAE values than either benchmark model. The regression results provide a more nuanced interpretation. While the federal funds rate exhibits a statistically significant relationship with future inflation, the communication measure is not statistically significant in the baseline linear forecasting specification. This finding suggests that much of the information contained in Federal Reserve communications may overlap with information already reflected in observable macroeconomic variables. Consequently, communication effects may not operate through a simple linear channel.

Nevertheless, improvements in out-of-sample forecasting performance indicate that Federal Reserve communications may still contain useful expectation-relevant information. One interpretation is that policy communications summarize policymakers’ assessments of future risks, inflation persistence, and prospective policy actions. Because policymakers observe a broader information set than the public, communications may reveal information regarding future economic conditions that is not fully captured by contemporaneous macroeconomic indicators alone (Bernanke, 2004; Blinder, 1998; Hansen et al., 2018).

The results should therefore be interpreted as evidence that communication-based measures complement traditional macroeconomic information rather than replace it. This interpretation is consistent with theories of information frictions, rational inattention, and expectation management, which emphasize that agents process information imperfectly and rely on publicly available signals when forming beliefs about future economic conditions (Sims, 2003; Mankiw and Reis, 2002; Evans and Honkapohja, 2001; Coibon and Gorodnichenko, 2015).

Several limitations should be acknowledged. First, the empirical implementation relies on FinBERT sentiment extraction rather than a fully estimated multi-agent LLM forecasting system. Second, the sample period is relatively short and largely reflects the post-pandemic inflation episode. Third, the analysis focuses exclusively on FOMC statements and does not incorporate speeches, meeting minutes, or Beige Book releases. Future research could construct explicit household, firm, and policymaker LLM agents and evaluate whether richer communication-based expectation measures further improve forecasting performance. An additional limitation is the relatively small sample size (57 monthly observations), which limits statistical power and may reduce the precision of coefficient

estimates.

VI. Conclusion

This paper develops a framework for incorporating Federal Reserve communications into real-time inflation forecasting. Motivated by recent advances in large language models, expectation formation is modeled as an information-processing activity in which agents combine quantitative macroeconomic information with qualitative policy communications. Using monthly U.S. data from 2021–2025, the analysis combines inflation, unemployment, industrial production, interest rates, and textual information extracted from FOMC statements. Communication-based measures are constructed using FinBERT and incorporated into a forecasting framework that is evaluated against standard benchmark models. The empirical results indicate that communication-based forecasts outperform both random-walk and autoregressive benchmarks. Although the communication measure is not statistically significant in the baseline linear regression, forecasting performance improves when information extracted from Federal Reserve communications is incorporated into the prediction process. These findings are consistent with the view that central-bank communications contain useful information regarding future inflation developments. More broadly, the findings suggest that central-bank communication may be viewed as an information channel through which policymakers reveal assessments of future economic risks. As large language models become increasingly capable of processing both structured and unstructured information, integrating textual signals into macroeconomic forecasting may represent a promising direction for future monetary-policy analysis (Korinek, 2024, 2023; Horton, 2023). The framework developed here provides a foundation for future research that explicitly models households, firms, and policymakers as heterogeneous LLM-based agents.

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APPENDIX

A1. Additional Forecast Evaluation

The FinBERT-based forecasting model generally produces smaller forecast errors than the benchmark models, particularly during periods of elevated inflation volatility. This result is consistent with the forecasting-performance metrics reported in the main text.

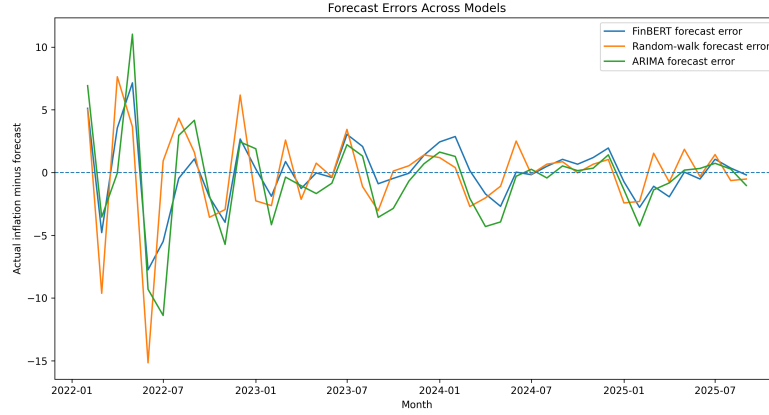


Figure 5: Forecast errors across models. Errors are defined as actual inflation minus predicted inflation. The FinBERT-based communication model generally exhibits smaller forecast errors than the benchmark forecasting models.

A2. FOMC Sentiment vs Inflation Scatter

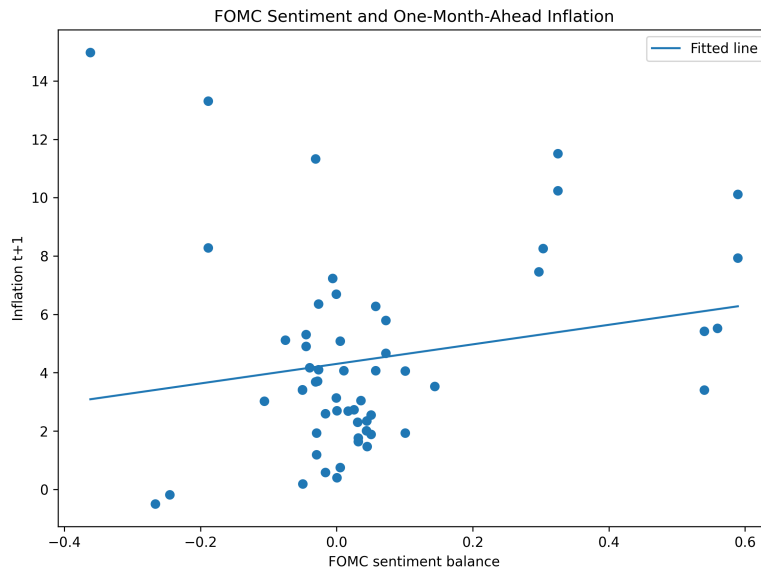


Figure 6: FOMC Sentiment vs Inflation Scatter

Figure 6 shows the relationship between FOMC communication sentiment and one-month-ahead inflation. While the linear association appears weak, the figure suggests that Federal Reserve communications may contain information that contributes to forecasting performance when combined with broader macroeconomic indicators.

A3. Transmission Channels in Household Decision-Making

This section illustrates how expected inflation, expected income, real wages, and expected real interest rates enter household decisions within the LLM-expectations framework.

a. Solving the Household Euler Equation

Household i maximizes

$$E_{i,t}^{LLM} \sum_{s=0}^{\infty} \beta^s \left[\frac{C_{i,t+s}^{1-\sigma}}{1-\sigma} - \frac{\varphi N_{i,t+s}^{1+\eta}}{1+\eta} \right], \quad (61)$$

subject to

$$P_t C_{i,t} + B_{i,t+1} = W_t N_{i,t} + (1 + i_{t-1}) B_{i,t}. \quad (62)$$

Let $\lambda_{i,t}$ denote the multiplier on the household budget constraint. The Lagrangian is

$$\mathcal{L} = E_{i,t}^{LLM} \sum_{s=0}^{\infty} \beta^s \left[\frac{C_{i,t+s}^{1-\sigma}}{1-\sigma} - \frac{\varphi N_{i,t+s}^{1+\eta}}{1+\eta} + \lambda_{i,t+s} (W_{t+s} N_{i,t+s} + (1 + i_{t+s-1}) B_{i,t+s} - P_{t+s} C_{i,t+s} - B_{i,t+s+1}) \right]. \quad (63)$$

The first-order condition with respect to consumption is

$$C_{i,t}^{-\sigma} = \lambda_{i,t} P_t. \quad (64)$$

The first-order condition with respect to bonds is

$$\lambda_{i,t} = \beta E_{i,t}^{LLM} [\lambda_{i,t+1} (1 + i_t)]. \quad (65)$$

Using

$$\lambda_{i,t} = \frac{C_{i,t}^{-\sigma}}{P_t}, \quad (66)$$

and

$$\lambda_{i,t+1} = \frac{C_{i,t+1}^{-\sigma}}{P_{t+1}}, \quad (67)$$

the bond condition becomes

$$\frac{C_{i,t}^{-\sigma}}{P_t} = \beta E_{i,t}^{LLM} \left[\frac{C_{i,t+1}^{-\sigma}}{P_{t+1}} (1 + i_t) \right]. \quad (68)$$

Multiplying both sides by P_t gives

$$C_{i,t}^{-\sigma} = \beta E_{i,t}^{LLM} \left[C_{i,t+1}^{-\sigma} (1 + i_t) \frac{P_t}{P_{t+1}} \right]. \quad (69)$$

Since

$$1 + \pi_{t+1} = \frac{P_{t+1}}{P_t}, \quad (70)$$

we obtain

$$\frac{P_t}{P_{t+1}} = \frac{1}{1 + \pi_{t+1}}. \quad (71)$$

Therefore, the Euler equation is

$$C_{i,t}^{-\sigma} = \beta E_{i,t}^{LLM} \left[C_{i,t+1}^{-\sigma} \frac{1 + i_t}{1 + \pi_{t+1}} \right]. \quad (72)$$

This expression shows that expected inflation enters household consumption decisions through the expected real return on saving.

The expected real interest rate is approximated by

$$r_{i,t}^e = i_t - E_{H,i,t}^{LLM}(\pi_{t+1}). \quad (73)$$

Thus, monetary policy affects consumption through

$$i_t \rightarrow r_{i,t}^e \rightarrow C_{i,t}. \quad (74)$$

Expected inflation affects consumption through

$$E_{i,t}^{LLM}(\pi_{t+1}) \rightarrow r_{i,t}^e \rightarrow C_{i,t}. \quad (75)$$

A4. Model Clarifications and Derivations

a. Derivation of the Labor-Supply Condition

The first-order condition with respect to labor supply is

$$-\varphi N_{i,t}^\eta + \lambda_t W_t = 0 \quad (76)$$

Using

$$\lambda_t = \frac{C_{i,t}^{-\sigma}}{P_t}, \quad (77)$$

gives

$$\varphi N_{i,t}^\eta = C_{i,t}^{-\sigma} \frac{W_t}{P_t}. \quad (78)$$

The labor-supply condition equates the marginal disutility of supplying labor to the marginal utility value of labor income.

b. Derivation of Firm Demand and Marginal Cost

Final output is produced using the Dixit–Stiglitz aggregator

$$Y_t = \left[\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (79)$$

Cost minimization implies the demand function

$$Y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t. \quad (80)$$

Individual firms produce according to

$$Y_{j,t} = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}. \quad (81)$$

The firm solves

$$\min_{K_{j,t}, N_{j,t}} W_t N_{j,t} + R_t^K K_{j,t} \quad (82)$$

subject to

$$Y_{j,t} = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}. \quad (83)$$

The cost-minimization Lagrangian is

$$\mathcal{L} = W_t N_{j,t} + R_t^K K_{j,t} + \mu_t [Y_{j,t} - A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}]. \quad (84)$$

The first-order conditions imply

$$MC_{j,t} = \frac{1}{A_t} \left(\frac{W_t}{1-\alpha} \right)^{1-\alpha} \left(\frac{R_t^K}{\alpha} \right)^\alpha, \quad (85)$$

which is the real marginal cost used in the firm's pricing problem.

c. Derivation of Market-Clearing Prices

Firm-level demand is

$$Y_{j,t}^d = \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t. \quad \varepsilon > 1. \quad (86)$$

Firm-level supply is

$$Y_{j,t}^s = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}. \quad (87)$$

Market clearing requires

$$Y_{j,t}^d = Y_{j,t}^s. \quad (88)$$

Therefore,

$$\left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}. \quad (89)$$

Divide both sides by Y_t for the market-clearing relative price gives

$$\frac{P_{j,t}^*}{P_t} = \left(\frac{Y_t}{A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}} \right)^{1/\varepsilon}. \quad (90)$$

Raise both sides to power $-1/\varepsilon$:

$$\frac{P_{j,t}}{P_t} = \left(\frac{Y_t}{A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}} \right)^{-1/\varepsilon}. \quad (91)$$

Hence the market-clearing firm-level price is

$$P_{j,t}^* = P_t \left(\frac{Y_t}{A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha}} \right)^{1/\varepsilon}. \quad (92)$$

The corresponding market-clearing quantity is

$$Y_{j,t}^* = A_t K_{j,t}^\alpha N_{j,t}^{1-\alpha} = \left(\frac{P_{j,t}^*}{P_t} \right)^{-\varepsilon} Y_t. \quad (93)$$

A5. Why LLM Expectations Instead of Rational Expectations

In a rational-expectations equilibrium, agents' beliefs coincide with the true model-implied conditional expectation:

$$E_t^{RE}(x_{t+1}) = E(x_{t+1} \mid \Omega_t), \quad (94)$$

where Ω_t is the full information set implied by the structural model. This assumption is analytically convenient but strong. It requires agents to know the relevant model structure, process all available information, and form internally consistent forecasts.

This paper instead assumes that agents form expectations using LLM-based information processing for households, and firm as follow:

$$E_{i,t}^{LLM}(x_{t+1}) = f_H(\mathcal{I}_t, \chi_i; \theta_H), \quad (95)$$

$$E_{j,t}^{LLM}(x_{t+1}) = f_F(\mathcal{I}_t, \psi_j; \theta_F), \quad (96)$$

This assumption is motivated by three established literatures. First, rational-inattention models emphasize that agents face limits in processing information ([Sims, 2003](#)). Second, sticky-information models show that agents update information imperfectly and infrequently ([Mankiw and Reis, 2002](#)). Third, adaptive-learning models replace full knowledge of the true model with belief updating based on observed information ([Evans and Honkapohja, 2001](#)). The LLM expectation operator is therefore not an ad hoc replacement for economic theory; it is a computational representation of bounded, heterogeneous, text-based expectation formation.

The key distinction is

$$E_t^{RE}(\cdot) \neq E_t^{LLM}(\cdot). \quad (97)$$

The paper does not replace household or firm optimization. It replaces only the expectation-formation process.

A6. References for Model Components

The household Euler equation follows the intertemporal consumption literature (Hall, 1978; Sargent and Ljungqvist, 2018). The firm production, demand, marginal cost, and pricing environment follows the New Keynesian monopolistic-competition framework (Calvo, 1983; Woodford, 2003; Gali, 2015). The monetary-policy rule follows the Taylor-rule tradition (Taylor, 1993). The replacement of rational expectations with LLM-generated expectations is motivated by information frictions, sticky information, adaptive learning, and survey-expectations evidence (Sims, 2003; Mankiw and Reis, 2002; Evans and Honkapohja, 2001; Coibon and Gorodnichenko, 2015).

A7. Derivation of the LLM Inflation Forecast Aggregator

The model generates three expectation signals: household inflation expectations, firm inflation expectations, and Federal Reserve inflation expectations. Household-level expectations are aggregated as

$$\bar{\pi}_{H,t+1}^{LLM} = \int_0^1 E_{i,t}^{LLM}(\pi_{t+1}) di. \quad (98)$$

Firm-level inflation expectations are aggregated as

$$\bar{\pi}_{F,t+1}^{LLM} = \int_0^1 E_{j,t}^{LLM}(\pi_{t+1}) dj. \quad (99)$$

The Federal Reserve expectation is

$$E_{Fed,t}^{LLM}(\pi_{t+1}). \quad (100)$$

Let the model-implied inflation forecast be a convex combination of these three expectation signals:

$$\hat{\pi}_{t+1}^{LLM} = \omega_H \bar{\pi}_{H,t+1}^{LLM} + \omega_F \bar{\pi}_{F,t+1}^{LLM} + \omega_{Fed} E_{Fed,t}^{LLM}(\pi_{t+1}), \quad (101)$$

where the weights satisfy

$$\omega_H + \omega_F + \omega_{Fed} = 1, \quad \omega_H, \omega_F, \omega_{Fed} \geq 0. \quad (102)$$

The convexity restriction ensures that the aggregate forecast remains bounded within the range implied by household, firm, and Federal Reserve expectation signals.

In the baseline specification, the weights may be calibrated equally:

$$\omega_H = \omega_F = \omega_{Fed} = \frac{1}{3}. \quad (103)$$

Future research may estimate the aggregation weights using forecast-validation samples. Let

$$\mathbf{x}_t = \begin{bmatrix} \bar{\pi}_{H,t}^{LLM} \\ \bar{\pi}_{F,t}^{LLM} \\ E_{Fed,t-1}^{LLM}(\pi_t) \end{bmatrix}, \quad (104)$$

and let

$$\boldsymbol{\omega} = \begin{bmatrix} \omega_H \\ \omega_F \\ \omega_{Fed} \end{bmatrix}. \quad (105)$$

Then the forecast can be written as

$$\hat{\pi}_t^{LLM} = \mathbf{x}_t' \boldsymbol{\omega}. \quad (106)$$

The optimal forecast-combination weights solve

$$\min_{\boldsymbol{\omega}} \sum_{t=1}^T (\pi_t - \mathbf{x}_t' \boldsymbol{\omega})^2 \quad (107)$$

subject to

$$\mathbf{1}' \boldsymbol{\omega} = 1, \quad \boldsymbol{\omega} \geq 0. \quad (108)$$

This yields an empirically disciplineable inflation forecast that combines private-sector expectations and Federal Reserve expectations while preserving the structural interpretation of the model.

A8. Data Sources and Variable Construction

Variable	Source	Construction
Inflation (π_t)	FRED	Annualized monthly CPI inflation computed from CPIAUCSL.
Unemployment (u_t)	FRED	Civilian unemployment rate (UNRATE).
Federal Funds Rate (i_t)	FRED	Effective federal funds rate (FEDFUNDS).
Industrial Production Growth (g_t^{IP})	FRED	Monthly growth rate of industrial production (INDPRO).
FOMC Statements (z_t^{Fed})	Federal Reserve Board	Official FOMC policy statements released after scheduled meetings.
Positive Sentiment (P_t^+)	FinBERT	Probability that an FOMC statement is classified as positive.
Negative Sentiment (P_t^-)	FinBERT	Probability that an FOMC statement is classified as negative.
Neutral Sentiment (P_t^0)	FinBERT	Probability that an FOMC statement is classified as neutral.
Communication Signal (SB_t)	FinBERT	Sentiment-balance measure: $SB_t = P_t^+ - P_t^-$.
Hawkish Score (H_t)	Author Construction	Frequency of hawkish monetary-policy terms divided by total words in the statement.
Dovish Score (D_t)	Author Construction	Frequency of dovish monetary-policy terms divided by total words in the statement.
Forecast Target (π_{t+1})	Author Construction	One-month-ahead annualized CPI inflation.

Table 4: Data Sources and Variable Construction